The AURA Framework: How to Make the Web Speak

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ABSTRACT
Nowadays, the visual channel prevails on the Web as the main medium for communication, as it is used to convey content, functionality and navigation/interaction capabilities. However, recently, a new category of applications is emerging on Internet: Aural Applications. The main beneficiaries of this new kind of applications are people who, permanently or temporarily, cannot look at the screen to interact with the application itself. Current approaches for developing "aural applications" (such as speech markup languages, code optimizations for screen-readers) provide technical solutions only, missing to adequately support the overall aural design process. This paper introduces important design issues to consider when conceiving aural applications - especially when dealing with "aural" information architectures, navigation and interaction features – and presents the AURA Framework: a methodological and technological framework for designing and implementing *"aural applications"*.

Categories and Subject Descriptors
H.5.2. **[Information Interfaces and Presentation]:** User Interfaces – *Input devices and strategies, Interaction styles, Standardization.*
H.5.4. **[Information Interfaces and Presentation]:** Hypertext/Hypermedia – *Architectures, Navigation, User issues.*

General Terms
Design, Human Factors.

Keywords
Aural Pages Design, Aural Applications Design, Interaction Primitives, Reading Strategies.

1. INTRODUCTION
Websites are usually designed for being "looked at", and the visual interface is considered the most important (or the only) means of communication between a user and the application itself.

Web interfaces simultaneously convey messages of different kinds:

a. *Content*: images, graphics, texts, videos are all meant to be looked at (except audio commentary, of course). Different content chunks are displayed at the same time (e.g. in one page), and information is often highly redundant across pages (e.g. important links, said "landmarks", are displayed in every page). Moreover, often, the visual arrangement of the page adds specific meaning: the relevance of a piece of content, for example, can be conveyed by its positioning (e.g. top left, for western users, or center page), the color, the font (size and style), etc.

b. *Interface semantics:* important messages (besides content) are conveyed by various visual-based elements. Layout and graphics, for example, convey the distinction between navigation mechanisms (links and menus) and the content. Orientation (i.e. where the user is "located" and how s/he got there) is also communicated through proper visual clues.

c. *Navigation/Interaction:* the graphic interface also conveys and makes affordable the interaction/navigation mechanisms available to the user. Moving from one page to another is based on the activation of links, and the user should be able to visually identify links, guessing their meaning, and moving the pointer of the mouse directly over one of them.

There are at least three scenarios in which a user cannot rely on the visual channel to use a website[1] (or an interactive application, in general):

1. Visual impairment: the visual channel is completely ruled out (blind users) or severely affected (strong sight problems).

2. Small devices: users of palm-held devices, smart phones and mobile devices in general can only partially rely on the screen, being this latter quite small.

3. Reality-enhanced scenarios: these are situations (e.g. while driving a car, walking in a city or in a museum) in which looking at the screen is not desirable or possible, since it is dangerous, or it is preferable to look elsewhere (e.g. to the countryside, the buildings, the monuments, the paintings, etc.). The application should convey "messages" to the user and still let him/her focus also on the surrounding environment.

[1] Websites, web applications and interactive applications will be used in an interchangeable manner from now on in this paper.

In order to support the above scenarios, web applications should be mainly (if not exclusively) “aural” rather than “visual”. Aural web applications, therefore, should rely upon audio, as the prevalent (or exclusive) channel for delivering information, and should allow for effective interaction not based upon visual mechanism (such as direct pointing with the mouse, for example).

The rest of this paper is organized as follows. In Section 2 some related works concerning technologies and techniques for “auralization” are briefly discussed. Section 3 introduces important design issues which need to be coped when conceiving aural applications. Section 4 presents the AURA Framework to systematically cope with such design isues and Section 5 describes the software architecture of the Page Reader, a novel tool for enhancing effective auralization. In Section 6, we present a tangible case study on how the page reader works. Finally, Section 7 draws some conclusions and points to current and future research.

2. RELATED WORK

In a nutshell, allowing aural access to web applications could require adapting web contents to be delivered via the aural channel. This objective has been addressed for some years both in the research and in the industrial arena [1]. Many solutions have been developed using different technologies that have lead in turn to many different standards[2][3], being later proposed or enforced by various agencies [4][5].

2.1 Screen Readers

The World Wide Web Consortium (W3C) developed specific guidelines for web application accessibility. The most important set of recommendation is the Web Content Accessibility Guidelines [3] that explains how to create accessible web content.

Web pages that are compliant with the W3C guidelines are intended to be interpreted by screen readers (i.e. JAWS for Windows [6]) or aural browsers (pwWebSpeak [7]).

However, the use of screen reader to access web pages has two main drawbacks. The first one is that they provide output in a linear, either audio or Braille, format. Due to this linearity, it is much easier for (blind) users to get disoriented in a complex hypertextual environment. The second one is that screen reader base the presentation of contents on the underlying HTML code of web pages. However, the HTML language is intended for a graphical representation of contents and does not provide any specific support to the aural delivering of information.

2.2 Web Pages Analysis and Transcoding

The idea of analyzing web pages and transcoding them into a more specific language for aural delivering has been explored in many works [1][8][9].

In [8], parts of HTML documents deemed "interesting" are marked by hand and the corresponding XPath is used to extract the "interesting" blocks. In this approach, if the document organization changes, the annotation must be redone. It focuses on segmentation and annotation of a given page based on accessible layouts manually predefined (using an annotation editor). The goal is to transform a page to make it more accessible by screen readers.

Other researchers have proposed the idea of extracting relevant content using semantics, such as [9]. Here, single pieces of contents are identified by hand and a transcoding system to deliver web pages via different channels (WML, VoiceXML, etc.) is defined.

In [1], web pages are partitioned and labelled using automatic structural and semantic analysis techniques; concept hierarchies (or *aural trees*) are thus obtained from HTML documents. Aural trees are then used to generate VoiceXML documents.

2.3 Speech Markup Languages

Recently, the industry also focused on the development of implementation solutions to build audio interfaces.

Two main languages lead the market in the field of voice applications: VoiceXML [10] and SALT [11].

VoiceXML is a XML language broadly used in telephone applications such as call centers or voicemail. The technology is supported by enterprise industries such as IBM, Motorola, AT&T and Lucent Technologies that agreed to develop a common standard in 1999. VoiceXML has three main goals: 1) simplify the creation of web-based voice response services; b) enable the integration of voice access into existing intranet services; c) support such applications for new voice capable devices [12]. Voice applications can take advantage of the same application development and deployment paradigms that are commonly used by visual applications. As visual browsers interpret HTML documents, voice browsers interpret VoiceXML documents.

However, in spite of this potential, VoiceXML does not allow integrating the aural channel with the visual channel: adapting a current content to be aurally accessed through VoiceXML often means to develop from scratch a new application. In fact, speech user interfaces are significantly different from their visual counterparts [13] and manually maintaining a correspondence between HTML and VoiceXML documents is a too time-consuming and error-prone task [14].
Automatic transformation of HTML pages into VoiceXML has been attempted with good results [15][16] but one of the most successful result in trying to overcome the barrier between the visual and audio channel is XHTML+Voice Profile [17]. In this approach, Web pages developed in XHTML are enriched with VoiceXML, allowing - at the same time - visual and aural access. Multimodal browsers supporting the XHTML+Voice Profile are Netfront and Multimodal Opera Browser, currently developed by IBM and Opera Software [18].

SALT (Speech Application Language Tags) is an XML based markup language used in HTML and XHTML pages to add speech output and interaction capabilities to web based applications. It is designed to allow multimodal and telephony-enabled access to information, and supports various types of user input, such as speech, keyboard, mouse, keypad or stylus. SALT is developed by Microsoft, and plug-ins are available to enable MS Internet Explorer to access SALT-enabled applications. The markup language permits integration of speech with web pages, separation of the speech interface from business logic and data, and the support of a wide range of devices [19]. The principle of SALT design is that programmers can use web pages as model for the aural dialogue interaction, using the single page as the key element to design aural interaction [20]; unfortunately, this is also

its main drawback. In fact, modifying the graphical layout of web pages usually requires the adjustment of the corresponding aural counterpart. SALT is younger than VoiceXML and is currently not fully adopted in the market of Speech Applications.

These languages, which we have briefly discussed, are competing for offering solutions to implement aural applications (i.e. coding efficiently and effectively aural interaction designs). However, the adoption of a language or another cannot dictate *per sè* design solutions, i.e. how the user experience should be, in terms of information architecture, interaction and navigation capabilities.

The approach presented in this paper differs from the above mentioned implementation-oriented solutions in three important ways. First of all, we try to clearly decouple the graphical layout of the pages from the corresponding aural layout (unlike Screen Readers, SALT and XHTML+Voice Profile). Secondly, we identify the single pieces of content by hand but we do not make use of XPath to extract the "interesting" information. This makes our approach more independent from organizational changes. Finally, we propose a mixed solution for content delivery. A*ural trees* are firstly generated from a web page, which, in its turn, can be either transcoded into a more specific language (i.e. VoiceXML) or delivered by means of the page reader (an *ad hoc* tool that we have developed).

3. AURAL PAGES: DESIGN ISSUES

By aural pages we mean web pages which have to be consumed via the aural channel. However, designing aural pages need to take into account the differences among the “aural” and the “graphical” layout of the pages themselves. These differences affect both the access and the navigation within them.

One of the main differences between the graphical and the aural layout is that while the first one allows the users to access the pages and to navigate within them in a random fashion, the second one limits the access and the navigation to be strictly sequential. It follows that, assuring the accessibility and the effectiveness of aural pages (and of the overall aural hypertext) requires a carefully design of the corresponding aural layout.

The following list summarizes the design issues to take into account when dealing with aural pages: they are not exhaustive but they provide a good starting point:

- *Reading orde*r: information should be presented to the user according to its importance within the page (starting from the most important one). The most important information is presented as first one; after that, the user can either continue listening the page or going ahead to surf the website;
- *Page structure*: users need to understand the structure of the aural page they access in order to easily find interesting information;
- *Meta-information vs. proper content*: the user should be able to distinguish between meta-information - that can be related either to the current page structure or to the website structure - and the proper content. On one hand, meta-information answers to questions such as "where are we now?", "what is the current list of items?", "what is the ordering criterion of items?" and so on. On the other hand, the proper content should correspond to information about a print, a painter, a drawing technique, etc. Furthermore, this allows the user to discern what information has to be read, reducing the overall page reading time.
- *Identifying the elements within the page*: it refers to the user's ability to clear identify different elements within the current page. Each element could be either a link or a piece of content; it can belong to a content or navigation section, etc. The user needs to clearly recognize it. This aspect is interrelated both to the page structure and to the distinction of the information type.
- *Moving around the aural page*: it refers to the user's ability to “interact” with the aural page. The user should be able to listen again a previously accessed content, to skip the listening of a content, to follow a link and so on.

In order to address these requirements in a systematic fashion, the AURA Framework has been developed. AURA offers a set basic primitives, notation tools and guidelines to support the design of aural interactions.

4. THE AURA FRAMEWORK

4.1 The AURA Model

The AURA model aims at providing a methodological instrument to describe the aural structure of web page. It allows both distinguishing between different types of contents (and links) within the web page and (re)organizing them in a more effective way for an aural consumption. It also provides a “device” independent representation of contents. Figure 1 shows the abstract syntax of the AURA Model.

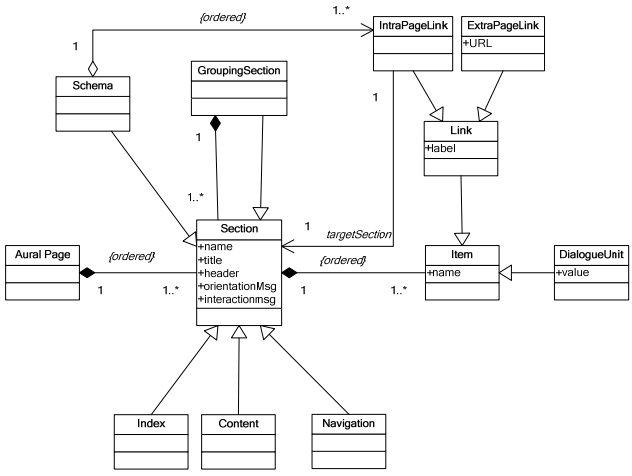

Figure 1. The AURA Model

The root of the model is the *Aural Page.* An Aural Page is composed of one or more objects of type *Section*. A *Section* represents an aural portion of the page and it is composed of one or more objects of type *Item*. Attributes such as “Title”, the “Header”, the “Interaction Message” and the “Orientation Message” also characterize each section. The Header provides a description of the contents within the section, the Interaction Message conveys the user information about the possible interaction mechanisms (i.e. with a long list of items) while the Orientation Message conveys the user information about his/her current position within the information architecture, about the number of links within a section and so on. Note that they provide

the user with a number of meta-information which, in the case of standard web pages, are usually conveyed by means of the graphical layout.

A Section, in turn, can be specialized according to its purpose:

- *Content section*: it contains the proper content that should be conveyed to the user;
- *Index section*: it introduces a group of content objects, enables to access this content objects and it may explain the meaning of the group itself (e.g. the list of books belonging to the same category).
- *Navigation section*: it groups a set of links that performs the same function (i.e. a main navigation bar in a website).
- *Grouping Section*: it groups together a set of related Sections.
- *Schema Section*: it allows the user to both easily identify the section composing the aural page and quickly access the content s/he is interested. It is composed of one or more objects of type IntraPageLink (each IntraPageLink will refer to a Section). We refer to the schema section as a meta-section because it does not provide information about the content of the section but about the overall aural page.

An *Item* identifies a single piece of content to be (dis)played within the aural page and it can be a *Link* or a *DialogueUnit*.

A *Link* is a reference and can be specialized as *ExtraPageLink* or *IntraPageLink*. An *IntraPageLink* refers to a *Section* and, usually, belongs to a *Schema Section*. An *ExtraPageLink* is characterized by an URL that refers to the (X)HTML page to load when selected.

A *DialogueUnit* is the smallest unit of information that is communicated to the end user.

The last step in defining the aural structure consists in specifying the mapping between the leaf elements of the aural page – Items and DiaolgueUnits – and the HTML elements belonging to the source page. Note that the mapping that we implemented is independent from the HTML code of the source page. In fact, we make use of a couple of HTML comments to identify the content components of the web page. For instance, Listing 1 shows an example on how it is possible to mark a link and an image within a standard web page.

```
…
<!-- BEGIN: id="LandmarkLink" -->
 <a                              href="Landmark.jsp"
class="MyStyle">Landmarks</a>
<!-- END: id="LandmarkLink" -->
…
…
<!-- BEGIN: id="PrintImage" -->
    <img alt="Woman in the Summernight"
    src="immages/big/big_stimme.jpg">
<!-- END: id="PrintImage" -->
…
```

Listing 1. Marking web contents: an example

However, the AURA model just provides a set of design primitives allowing to define the aural structure of web pages but it does not provide any kind of information on how the user will effectively access the page contents (reading strategies) and on how s/he can interact with the "aural page" (interaction primitives).

4.2 The Reading Strategy

A reading strategy represents the ways in which a user can listen to an aural page. Whereas for the visual interaction the reading strategy is entirely visually-driven (the user quickly scans the page, then s/he focussed on the content items, or on the links, and so on), aural design needs to define as well reading strategies which can effectively support the interaction.

Defining the reading strategy for an aural page consists in specifying:

1. the reading order of items within sections (e.g. within the content section, the main header, then the content text, then the texts describing the pictures);
2. the reading order of sections within the aural page (e.g. the content section first, then the local navigation section, then the main navigation section, and so on);
3. the reading mode of each section: it consists in specifying whether the content belonging to the section has to be read the first time the use will access the section, every time or never. Some possible access modalities are:
   - *Always*: the section is read every time the user accesses the page (i.e. for content sections).
   - *Only the first*: the section is read only the first time the user accesses the page. The next times only an alternative header will be read. The overall content of the section will be read only after an explicit user's request (i.e. for the schema section).
   - *On demand*: only the header of the section is read. The overall content of the section will be read only on explicit user's request (e.g. navigation sections may be read only when needed).

   Note that different access modalities for sections allow speeding up the reading time of the aural page and entail a shift of paradigm (with respect to current screen reader usage): the user has not to choose which sections to skip but she explicitly selects which sections to read;
4. different navigational paths among sections: it consists in breaking up the linearity of the fruition imposed by the aural channel, allowing the user to directly access a section of interest at any time (by means of IntraPageLinks).

4.3 Interaction Primitives

"Point and click" is the simple action of a user moving a cursor over a given location on a screen (point) and then clicking a mouse button or other pointing device.

However, the characteristics of the aural channel make the "point and click" paradigm not sufficient to assure an effective user-experience; it would force the user to sequentially listen to the page contents and links and allowing the interaction only when a link is read. To overcome this situation, at least the following interaction primitives need to be considered:

1. Go to the Next/Previous Section: they allow the user to move forward and backward across page sections;

2. Go to the Next/Previous Item: they allow the user to move forward and backward across items within a section (i.e. to skip a link or to listen again a piece of content);
3. Go to the Schema Section: it provides a direct access mechanisms to the section (i.e. to recover the orientation within the page or to select another section to listen);
4. Go Back to the Last Accessed Section: it provides a back functionality within the aural page (i.e. after accessing the Schema Section it is possible to go back to the last accessed content section without selecting the corresponding link);
5. Follow the link: it is the simplest action primitive and can be activated only when a link is read. Note that the result of following the link depends both on the type (the link can be within the same aural page or to another one) and on target reference of the link itself;
6. Pause/Resume: it allows to stop/resume the reading of the current item;
7. Reload Section: it allows starting over the reading of the current section.

What is important to note is that the interaction primitives are easy to learn - just 10 interaction primitives - and to use, requiring no special knowledge to the user.

Furthermore, interaction primitives only define the "communication language" between the user and the aural application: they do not provide any kind of information on how the communication happens. To support the communication one or more interaction devices should be conceived.

5. THE PAGE READER ARCHITECTURE

Figure 2 shows the software architecture of Page Reader which can be logically divided into three main layers: Data source, Core and Presentation/Interaction.

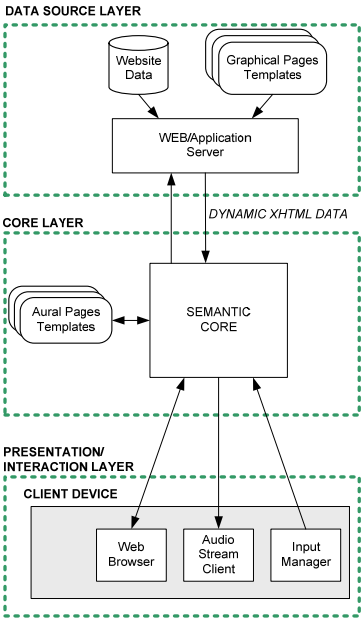

Figure 2. The PAGE READER architecture

5.1 Data Source Layer

This layer is composed of the web application to be aurally rendered. It is accessed via network and usually resides on the customer webserver. The server offers XHTML pages enriched with AURA tags to the *core layer* upon request.

5.2 Core Layer

The core owns the representation of all semantic structures in a data structure named *Aural Tree*. This data structure is used to allow stateful navigation within the page. Upon client device request the core retrieves dynamic data from the data source layer, fills an instance of the aural structure and outputs coherent data to the *presentation/interaction layer*. The core handles multiple channels in order to provide end users with all needed streams, such as audio stream, interaction components and visual XHTML data. Source page caching techniques are used in order to reduce latency time induced by data retrieval from the data source layer, still guaranteeing full dynamism.

The core includes five main components in order to interact with user devices:

- an *audio streaming server* such as Flash Media Server [21] or any other streaming technology [22];
- a *Text-To-Speech engine* that transforms textual data obtained from the source website and additional metadata (such as navigational hints) into formats handled by the Audio Streaming Server;
- a *visual proxy* that forwards the XHTML and enriches the pages with needed plug-ins such as stream clients and user input handlers such as Java™ applets or ActiveX™ components;
- a *user input manager* server that receives and dispatches all inputs coming from the end user devices;
- a *semantic primitives mapping register*, used to map input interrupts into coherent semantic commands to be run on the semantic tree.

5.3 Presentation/Interaction Layer

This layer enables user interaction with the AURA architecture and includes the user interface components. These components reside on the client side (Desktop, PDA, Cell phone) and are device dependent.

It is composed of:

- a *standard web browser* that supports all needed plug-ins technology such as audio stream clients or Java™ JRE;
- an *audio stream client* capable of simple actions to control the streamed data such as play, stop, pause and resume. The end user does not directly control the stream client; all needed actions are handled by the client input manager. Many technologies are available: Flash Player, Real Player, Media player, Java Applets or ActiveX components;
- an *input manager* that is capable to handle user inputs, such as keyboard strokes or touchscreen events and to change the browser status by loading a new page. The manager sends

all received commands to the core layer that does all needed operations to provide user with requested data. The core send back an immediate audio feedback response for any received input to the client, in order to enhance the navigation experience.

The interaction layer is the only interface exposed to end-users by the AURA Framework.

6. THE FRAMEWORK AT WORK

In order to fix the AURA Framework to a tangible case study, in this section we will show in practice how to define the aural structure for a sample page belonging to a real website and how the aural page will be vocally rendered by the Page Reader. The page shown in Figure 3 refers to the description of a print of Edvard Munch, from the www.munchundberlin.org website, aiming at promoting a temporary exhibition of Munch's prints at the State Museum in Berlin.

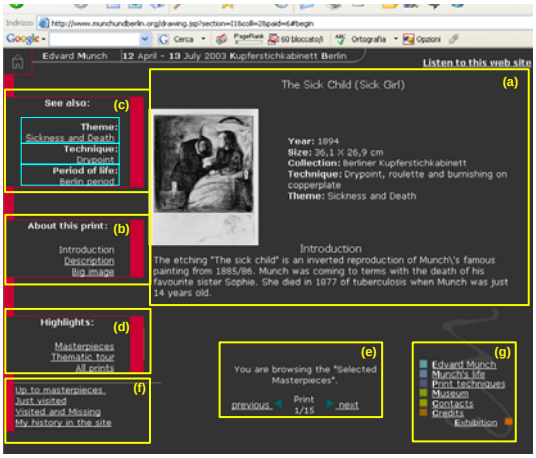

Figure 3. The page of a print from www.munchundberlin.org.

The page is (logically) interpreted as composed of seven sections, from (a) to (g):

(a) This section contains the actual content about the print (it is a *content section*). Contents about the print are structured into a Title, a set of Technical Data, a set of Descriptive Data and an Image (see Figure 4);

(b) This section contains a set of links to additional information about the current print, such as a detailed Description or a Big Image (see Figure 4). It is a *navigation section*;

(c) This section contains a set of links to other contents related to the print, it is an *index section*. Notice that the section is composed of three sub-sections, it is also a *grouping section*;

(d) This section contains a set of links to highlights, it is an *index section*;

(e) This section contains information about the group of subjects from which the user accessed the print (i.e. the "masterpieces" of Munch), and about the current position of the print within this group. It also provides the user with links to browse the print in the group. It is an *orientation (navigation) section*;

(f) the section contains a set of service links, it is a local *landmark (navigation) section*;

(g) This section contains a set of general links, it is a *landmark (navigation) section* (see Figure 4).

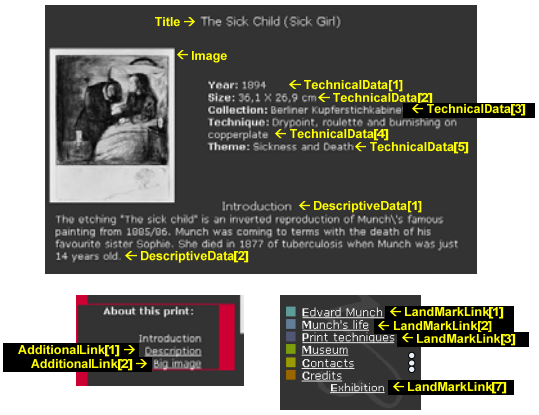

Figure 4. Contents and Links within the print page.

Let us assume now a set of important user requirements (which are however quite general and reasonable for many applications):

- the main user's interest is about the proper content related to the print;
- the user must be able to easily deepen his/her knowledge about the print (by accessing to related contents);
- landmark sections are boring, the user will access them only after an explicit request;

These assumptions provide the input to devise a suitable and flexible reading strategy (see Section 4.2).

Figure 5 shows the aural structure related to the print page.

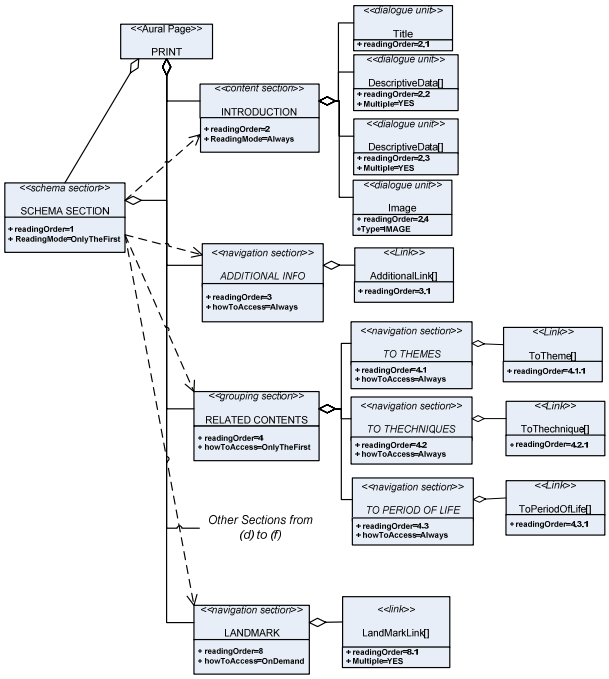

Figure 5. The aural structure for the print page.

Note that, Figure 5 just shows the aural structure of a page: in other words an *aural template*. All the prints of the website will be (dis)played according to this template. For instance, Table 1 presents the transcription of what the user will hear if s/he will access the print – The Sick Child – shown in Figure 3 in sequential order.

Table 1. Transcription of the print :"The Sick Child". Bold texts refer to meta-information

TRANSCRIPTION	AURAL ELEMENT
- **This page contains the introduction about the print:** The Sick Child (Sick Girl)	Aural Page: Header
- **This page is structured into 7 sections.**	SchemaSection: Header
- **You can either select one of them to start with or wait for the first one. Sections are:**	SchemaSection: Interaction message
- **Section 1: Actual content** - **Section 2: Additional Information** - **Section 3: Related Contents** - **Section 4: Orientation information** - **Section 5: Highlights** - **Section 6: Service Links** - **Section 7: Landmarks**	SchemaSection: Links
- **Actual content** - **The section contains information about the print**: The Sick Child (Sick Girl)	ContentSection: title and header
- Introduction - The etching "The sick child" is an inverted reproduction of Munch's famous painting from 1885/86. … - **Beginning of Technical Data:** - Year: 1894. - Size: 36,1 X 26,9 cm. - Collection: Berliner Kupferstichkabinett. - Technique: Drypoint, roulette and burnishing on copperplate. - Theme: Sickness and Death. - **End of Technical Data.** - **Beginning of the Image Description:** - … *Alternative text for the image* - **End of the Image Description.**	ContentSection: dialogue units
- **Additional information** - **The section contains a set of links to additional information about the print:** The Sick Child (Sick Girl)	StructuralLinksSection: title and header
- **Link 1 of 3:** Introduction. - **Link 2 of 3:** Description. - **Link 3 of 3:** Big image	StructuralLinksSection: links
- **Related contents** - **The section contains a set of links to other contents related to the print:** The Sick Child (Sick Girl)	IndexSection: title and header
- Theme: - **Link 1 di 1:** Sickness and Death.	ThemeSection: header and links
- Technique: - **Link 1 of 1:** Drypoint.	TechniqueSection: header and links
- Period of life: - **Link 1 of 1:** Berlin period.	PeriodOfLifeSection: header and links
- **Orientation information** - **The section contains information about the current collection and the position of the print within the collection.** - **You can select a link to go to the previous or next print.**	OrientationSection: title, header and interaction message
- You are browsing the "Selected Masterpieces" - **Link:** Go to the Previous print. - **Link:** Go to the next print.	OrientationSection: dialogue unit and links
- **Highlights** - **The section contains 3 links to Highlights**	HighlightsLinksSection: title and header
- **Link 1 of 3:** Masterpieces. - **Link 2 of 3:** Thematic tour. - **Link 3 of 3:** All prints.	HighlightsLinksSection: links
- **Service Links** - **The section contains 4 Service links**	ServiceLinksSection: title and header
- **Link 1 of 4:** Up to masterpieces . - **Link 2 of 4:** Just visited. - **Link 3 of 4:** Visited and Missing. - **Link 4 of 4:** My history in the site	ServiceLinksSection: links
- **Landmarks** - **The section contains 7 landmark links**	LandmarkSection: title and header
- **Link 1 of 7:** Edvard Munch. - **Link 2 of 7:** Munch's life - **Link 3 of 7:** Print techniques - **Link 4 of 7:** Museum - **Link 5 of 7:** Contacts - **Link 6 of 7:** Credits - **Link 7 of 7:** Exibition	LandmarkSection: links.

Thanks to the set of interaction primitives supported by AURA, the user can interact with the website as shown in the example of Figure 6.

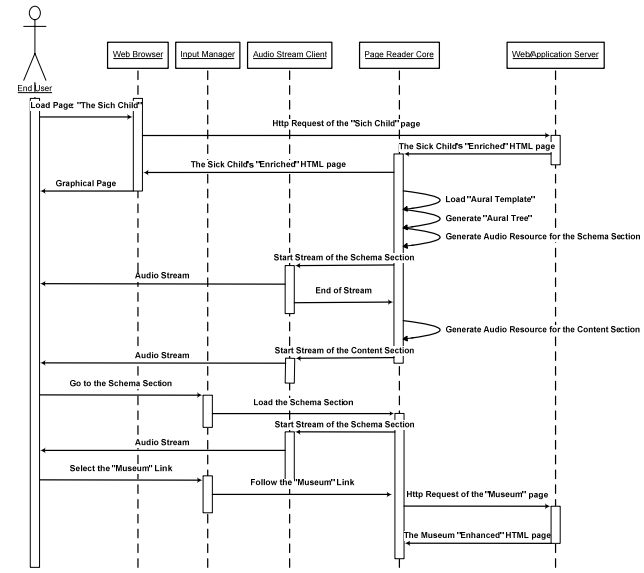

Figure 6. An example of interaction between the end user and the AURA architecture

At first, s/he will listen to the schema section and the content section. After that, s/he will go back to the schema section and will continue his/her navigation by selecting the Landmark Section. The user will end his/her navigation within the current page by selecting the "Museum" link.

7. CONCLUSION AND FUTURE WORK

In this paper, we have presented and partially analyzed important design issues to be considered when conceiving aural applications

and we have proposed a methodological and technological solution.

Specifically, the methodological solution consists in separating the aural structure of web pages (the AURA Model) from the ways in which a user can both listen (the reading strategy) and access them (the interaction primitives). On the other hand, the PAGE READER provides the technological support to deliver aural applications.

The main advantages of our approach are a limited, non invasive approach to original code of web pages, separation between the graphical and aural layout, and an automatic multi-channel deployment via a network proxy service (programmer point of view). Moreover, end users can seamlessly access the same content via different channels using a common set of interaction primitives, allowing a reduced learning curve.

Current and future research is focusing on the development and refinement of the solutions proposed in this work, and by validating their effectiveness on a larger basis of empirical testing.

Moreover, thanks to the ORA Project [23], a re-interpretation of existing design models and methods in an aural perspective is on going. The expected result is an extension of design methodologies aimed at supporting the design of aural applications.

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